

●●● C source code

```
1 #include <stdio.h>
2 int main(){
3     int n,a[100],f[100],g[100],st[100],top=-1;
4     scanf("%d",&n);
5     for(int i=0;i<n;i++)scanf("%d",&a[i]);
6     for(int i=n-1;i>=0;i--){
7         while(top>=0&&a[st[top]]<=a[i])top--;
8         f[i]=top<0?-1:st[top];
9         st[++top]=i;
10    }
11    top=-1;
12    for(int i=n-1;i>=0;i--){
13        while(top>=0&&a[st[top]]>=a[i])top--;
14        g[i]=top<0?-1:st[top];
15        st[++top]=i;
16    }
17    for(int i=0;i<n;i++)printf("%d%c",f[i]<0||g[f[i]]<0?-1:g[f[i]]+1,i==n-1?'\n':' ');
18    return 0;
19 }
```

●●● Program output

```
-1 4 -1 -1 -1
```